

Machine learning for requirements engineering (ML4RE): A systematic literature review complemented by practitioners' voices from Stack Overflow

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ABSTRACT

Context: The research of machine learning for requirements engineering (ML4RE) has attracted more and more attention from researchers and practitioners. Although pioneering research has shown the potential of using ML techniques to improve RE practices, there lacks a systematic and comprehensive literature review in academia that integrates an industrial perspective. Specifically, none of the reviews available in ML4RE have considered the grey literature, which is primarily from practitioner origin and is more reflective of the real issues and challenges faced in practice.

Objective: In this paper, we conduct a systematic survey of academic publications in ML4RE and complement it with the practitioners' voices from Stack Overflow to complete a comprehensive literature review. Our research objective is to provide a comprehensive view of the current research progress in ML4RE, present the main questions and challenges faced in RE practice, understand the gap between research and practice, and provide our insights into how the RE academic domain can pragmatically develop in the future.

Method: We systematically investigated 207 academic papers on ML4RE from 2010 to 2022, along with 375 questions related to RE practices on Stack Overflow and their corresponding answers. Our analysis encompassed their trends, focused RE activities and tasks, employed solutions, and associated data. Finally, we conducted a joint analysis, contrasting the outcomes of both parts.

Results: Based on the statistical results from collected literature, we summarize an academic roadmap and analyse the disparities, offering research recommendations. Our suggestions include the development of intelligent question-answering assistants employing large language models, the integration of machine learning into industrial tools, and the promotion of collaboration between academia and industry.

Conclusion: This study contributes by providing a holistic view of ML4RE, delineating disparities between research and practice, and proposing pragmatic suggestions to bridge the academia-industry gap.

1. Introduction

Requirements engineering (RE) is an imperative step in the early stage of software engineering (SE). Although researchers have been continuously investigating methods and techniques to facilitate requirements practices, the entire process of requirements engineering still requires a large amount of manual effort (e.g., eliciting stakeholder requirements via interview or classifying requirements according to certain requirements taxonomy). A primary reason for this is that RE activities typically require in-depth domain knowledge and advanced analysis skills, which is difficult to be fully automated.

In recent years, the rapid development of machine learning (ML) technology has been propelled by improvements in computational power. The successful applications of ML in areas such as natural

language processing, image recognition, and data mining have provided opportunities to apply ML techniques in the field of RE. The application of ML technology in RE offers a smarter and more efficient approach to handling requirement data. For example, ML can assist in automatically categorizing requirements since it can summarize potential requirement information. Furthermore, with the rapid development of information techniques, our daily work and life are being digitalized. As a result, more and more requirements-related data have been digitized and publicly available, promoting the research of machine learning for requirements engineering (ML4RE). For example, user reviews of mobile applications have been intensively investigated to mine users' requirements [1].

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Previous research has explored the ML4RE. Iqbal et al. [2] conducted a survey to obtain an overview of how ML techniques can help with RE activities. Additionally, Zamani et al. [3] performed a mapping study of ML applications in RE, analysing 65 papers to assess ML's effectiveness in automating RE tasks. Their works focused on the entire RE process, shedding light on how ML techniques influence and streamline various phases. Besides, some research delves into specific RE activities or tasks. For instance, Lim et al. [4] reviewed current state-of-the-art approaches to data-driven requirements elicitation from dynamic data sources.

We notice that these studies have predominantly focused on academic publications and missed integrating insights from grey literature sources, such as industry blogs and forums. In contrast to academic publications that are mainly published by researchers, grey literature is constantly produced by practitioners on a great scale, shedding light on the "state of the practice" [5]. As indicated in [6], integrating grey literature into systematic literature reviews can bridge the gap between academic research and industry practices, offering a more holistic view of challenges and solutions. Although there is an increasing number of systematic literature reviews considering grey literature in SE studies [7,8], they are not as prevalent in RE.

To fill the gap in the RE field, this paper aims to conduct a systematic literature review on ML4RE, integrated with insights from grey literature sourced from Stack Overflow. Specifically, our literature review consists of three parts. The **white part** involved a review of 207 papers published between 2010 and 2022, specifically focusing on ML4RE. Meanwhile, the **grey part** encompassed an analysis of 375 questions and their corresponding answers gathered from Stack Overflow discussions on RE practices during the same period. Lastly, our **joint analysis** involved contrasting the outcomes of these two parts to discern their similarities and differences. We analyse the literature's trends, RE activities, RE tasks, solutions, and data. Research results indicate that both exhibit substantial interest in RE analysis and requirements documentation. Beyond commonalities, the white part tends to focus on RE elicitation, while the grey part places greater emphasis on RE management. The white part underscores the use of ML technologies such as SVM, CNN, and BERT. In contrast, the grey part relies more on tools such as Microsoft TFS, Jira, and IBM Rational DOORS, with ML techniques such as LDA, POS, TF-IDF, and neural networks. Additionally, the grey part pays particular attention to the user story and use case, areas relatively less explored in the white part. Based on these findings, we summarize an academic roadmap and conduct a detailed analysis of the disparities between the white part and the grey part. Subsequently, we propose research suggestions, advocating for the development of intelligent question-answering assistants using large language models and the integration of machine learning into industrial tools. We further encourage enhanced collaboration between academia and industry to gain a deeper understanding of real-world research problems and data.

In summary, this paper makes the following contributions. Firstly, we provide a comprehensive overview of the current state of ML4RE research. Secondly, we investigate the real-world circumstances of RE practitioners through grey literature obtained from Stack Overflow. Thirdly, we highlight the gap between research and practice in the ML4RE domain, particularly in areas less attended to by researchers. Lastly, to bridge the gap between industry and academia, we propose practical research suggestions in ML4RE.

In the remaining part of this paper, we present the related work in Section 2. Section 3 presents the research protocol for our systematic literature review. In the next three sections, we present the results of this survey and answer the research questions. Section 4 presents results from the white part, Section 5 focuses on the grey part, and Section 6 presents the outcomes of the joint analysis. Based on the results, we have some discussion and give several suggestions in Section 7. Section 8 includes the analysis of threats to the validity of this survey. Finally, we conclude the paper in Section 9.

2. Related work

The related work of our literature review in ML4RE comes from two sources. The first set of reviews pertains to ML4RE. The second set consists of systematic literature reviews in the field of SE. In the following two subsections, we will present the related work in detail from these two sources.

2.1. Literature reviews about ML4RE

In this section, we present a compilation of relevant literature reviews focusing on ML4RE. We have found several literature reviews, some addressing the entirety of the RE process, while others concentrate on specific RE activities or tasks. Therefore, this subsection is divided into two parts to introduce these works in detail.

2.1.1. Entire requirements engineering process

Two papers focus on the entire requirements engineering process and provide an overview of how machine learning techniques play a role in the various stages of requirements engineering. Iqbal et al. [2] surveyed many research papers to obtain an overview of how ML is helping RE. They observed the impact of ML in all five phases of RE and discussed the specific problems solved by ML, the features, ML algorithms, and datasets. However, the papers they surveyed were not obtained from a systematic search process, and therefore, the conclusions drawn may not be systematic and comprehensive. Zamani et al. [3] performed a mapping study of ML applications in RE, analysing 65 papers to assess ML's effectiveness in automating RE tasks. It identifies these studies' techniques, challenges, datasets, and evaluation metrics. Compared to our work's comprehensive review, this paper focuses more on the empirical aspects of ML in RE, offering specific insights into ML's practical efficacy in this field. Besides, Zhao et al. [9] focused on NLP4RE, analysing 404 studies to understand the application of NLP in RE. Because there are overlaps between NLP and ML, we regard this research as one of the related works to analyse.

2.1.2. Part of requirements engineering process

In the requirement elicitation process, G.C. Sampada et al. [10] provided an overview of different approaches to automate requirement elicitation and specification in the software development cycle. Lim et al. [4] reviewed the current state-of-the-art approaches to data-driven requirements elicitation from dynamic data sources and identified research gaps. Cheliger et al. [11] selected 86 papers to summarize and analyse studies incorporating ML and NLP into demand elicitation. They categorized the techniques for constructing ML-based requirement elicitation methods into five parts.

In the requirements classification process, Alrumaih et al. [12] presented some research studies on requirements classification and discussed limitations to yield suggestions for improvement. Perez et al. [13] identified the applications of ML techniques in classifying software requirements by 13 selected papers and summarized the most recurrent classification algorithms and the most frequent training datasets. Khe-lifa et al. [14] investigated the applicability of machine learning techniques in software requirements classification and requirements change requests classification. Additionally, Kadebu et al. [15] concentrated on security requirements engineering and explored the application of ML techniques in security requirements extraction and classification.

In the requirements management process, Xu et al. [16] summarized eight ML methods used in requirements management and determined 18 evaluation indexes for requirements management in the ML method. Their work serves as a foundational understanding of a wide array of ML techniques used in managing requirements, while some studies delve deeper into specific tasks. Achimugu et al. [17] reviewed techniques for prioritizing software requirements through 73 related articles and concluded several limitations in existing prioritization techniques. Moreover, Li et al. [18] conducted a systematic mapping study

with 26 studies and summarized 32 ML technologies for requirements traceability.

From the above, it is evident that there is a large amount of excellent literature review on the topic of ML4RE. However, there is currently a lack of research that considers the integration of the grey part, representing industrial aspects. Our work aims to bridge this gap by comprehensively considering academic and industrial perspectives.

2.2. Literature reviews in SE

In the field of SE, there is a plethora of systematic literature reviews. We have selected the relevant papers that integrate grey literature. We categorized them into three domains based on the paper topics, which will be introduced one by one below.

2.2.1. Methodologies in software development

Methodologies in software development are software engineering development models or value systems embraced by broad developers in the software development process. Well-known methodologies in software development include Agile, DevOps, and one brought up based on the latter, DevSecOps. França et al. [19] performed the literature review aiming at characterizing DevOps from multiple perspectives. Amaro et al. [20] aimed to clarify DevOps capabilities and their relationship with DevOps practices. With the development of DevOps, security is becoming increasingly important to software engineering. DevSecOps comes up with the integration of modernized security methods and DevOps. Myrbakken and Colomo-Palacios [8] conducted the literature review to give an overview of the definition, significance, benefits, and challenges of DevSecOps. For DevSecOps implementation quality, Prates et al. [21] performed the literature review to identify metrics teams can use to measure the effectiveness of DevSecOps methodology implementation inside organizations. More and more IT companies are adopting microservices architecture to deliver their business. Soldani et al. [22] systematically selected and analysed the industrial grey literature on the pains and gains of microservices. Bhandari and Colomo-Palacios [23] focused on holacracy for software development teams. Some works focus on how to use ML to assist DevOps in development. Recupito [24] conducted a multi-vocal literature review to explore MLOps tools and their functionalities in automating Machine Learning pipelines with DevOps practices.

2.2.2. General software engineering

Some research works have focused on adopting voices from practitioners in a more general software engineering field. Kamei et al. [25] conducted a tertiary study to understand how secondary studies use grey literature. Considering the state of researchers in software engineering were still struggling with research relevance and utility to practitioners, Garousi et al. [26] carried out the literature review. They found insights into the causes of low relevance and suggestions for improving it. Rainer and Williams [27] conducted a tertiary study on the studies empirically investigating software practice using blog-like documents. Alves et al. [28] presented a comprehensive taxonomy of practices used in the industry for building machine learning systems, offering valuable insights to organizations for improving and managing their ML processes and practices. Heiland et al. [29] provided an overview of design patterns for AI-based systems, including both new and adapted patterns, collected through a multi-vocal literature review.

2.2.3. Specific parts in software engineering

Literature reviews exist in various parts of SE. In the domain of software testing, Raulamo-Jurvanen et al. [30] carried out a grey literature review to identify the potential pitfalls of existing processes and opportunities for comprehensive tool evaluations. Still, for test automation, Garousi and Mäntylä [7] performed the literature review on when and what to automate in software testing. Garousi et al. [31] focused on test maturity assessment and test process improvement and

conducted the literature review. Garousi and Küçük [32] performed a multi-vocal literature mapping on test smells in both the scientific and grey literature. Felderer and Garousi [33] surveyed software testing in both industry and academia and they proposed their reflections on improving the link and synergies between industry and academia in software testing. Regarding the RE domain, Tripathi et al. [34] used the literature review to find scientific and grey literature. They explored how software startups used RE elicitation, documentation, prioritization, and validation.

Overall, the landscape of SE has seen substantial developments in the systematic literature reviews, with numerous excellent works combining perspectives from grey literature for analysis. However, our analysis reveals that among these works, they have yet to delve into the ML4RE, while our work fills this gap.

3. Research protocol

We applied the standard protocol structure of systematic mapping study in software engineering developed by Kitchenham et al. [35]. Our study consisted of a **white part** and a **grey part**, each reviewed separately. We then developed a **joint part** to combine findings from both parts.

3.1. Objectives and research questions

In particular, we have used the Goal-Question-Metric methodology [36]. Our objectives encompass a thorough examination of the ML4RE research landscape. Additionally, we seek to investigate practical needs in RE and methods in ML drawn from grey literature. Finally, we compare academic research's focus on RE in the problem domain and the methods employed in the solution domain with real-world practices in Stack Overflow. These objectives lead to the formulation of research questions (RQs).

RQ1: What is the current state of academic research in ML4RE?

RQ2: What is the current state of ML4RE practices in Stack Overflow?

RQ3: How do Stack Overflow insights on ML4RE align with or differ from academic research findings?

In order to answer these RQs, we define a series of tags as metrics to specify the data or information that needs to be extracted from papers. In the subsequent sections, we detail the literature search processes for the white and grey parts. We also explain the metrics used to answer the RQs and the data extraction and synthesis process.

3.2. White part selection process

Fig. 1 presents the procedure of the selection process in the white part, and the number of papers derived at each step is also in the figure. The following subsections present the details of each step.

3.2.1. Systematic search

We conduct our white part systematic search using the pre-defined search string in four selected academic databases. The systematic search process in the white part includes three steps. We explain each step in the order shown in Fig. 1.

Search sources selection. The search sources can significantly influence the quality of the systematic literature review. As suggested by many studies [35,37], search sources should include multiple authoritative databases. Specifically, we select four academic databases: IEEE Xplore, ACM Digital Library, SpringerLink, and Scopus. These four databases can cover the majority of the literature related to requirements engineering [38].

Search string definition. The essential theme of our study is machine learning for requirements engineering. We divide the search string into two parts: the RE part and the ML technique part. These two parts are connected using the "AND" operator. Specifically, each part

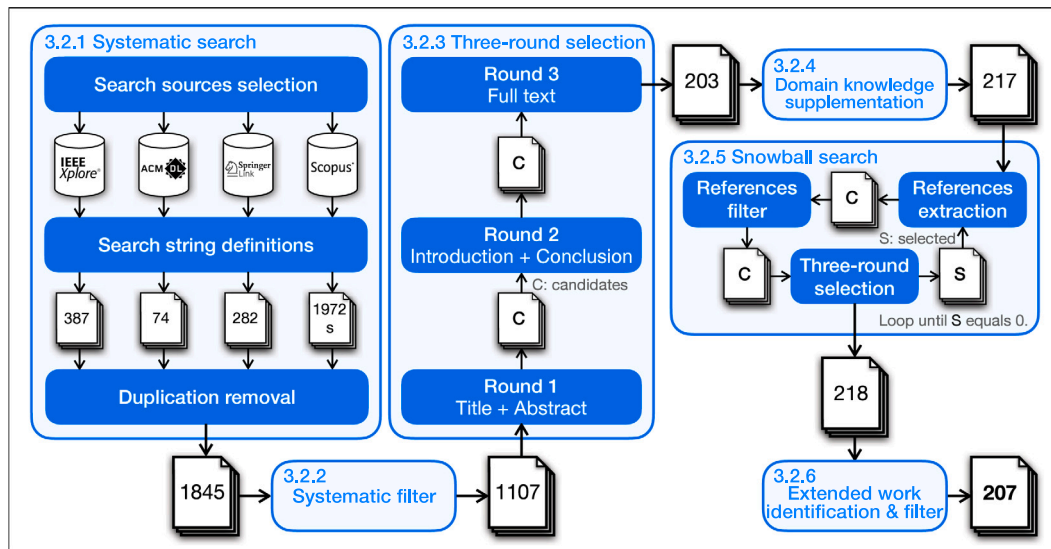


Fig. 1. The selection process of the white part papers.

Table 1
The search string used in the white part.

Category	Keywords	
Requirements Engineering	requirements engineering	requirements specification
	requirements elicitation	requirements validation
	requirements analysis	requirements management
	software requirements	quality requirements
	functional requirements	non-functional requirements
Machine Learning	machine learning	semi-supervised learning
	machine-learning	deep learning
	supervised learning	reinforcement learning
	unsupervised learning	bayesian learning
	neural network	support vector machine
	decision tree	bayesian network

encompasses relevant key terms, and the key terms within each part are connected with the “OR” operator. To ensure comprehensive search coverage and minimize manual selection efforts, several iterations were conducted to determine and refine the keywords. The complete search terms are detailed in Table 1. Additionally, a snowballing search in Section 3.2.5 will be employed to expand the scope of the search further.

Duplication removal. We search for papers whose titles, abstracts, or keywords contain the pre-defined search string in four databases and confine the publication year between 2010 and 2022. It is worth noting that SpringerLink only supports full-text searches, necessitating additional checks of papers in this database’s search results. Specifically, we utilized functions in Google Sheets to concatenate the title, abstract, and keywords of each paper and matched them with predefined search strings using regular expressions. This process reduced the number of papers from SpringerLink from 7367 to 282. Subsequently, we obtained 387 papers from IEEE Xplore, 74 papers from ACM Digital Library, 282 papers from SpringerLink, and 1972 papers from Scopus. After removing all of the duplications, there are 1845 papers left.

3.2.2. Systematic filter

The initial search results from those academic search engines contain lots of unwanted cases, such as keynotes, posters, or papers that are too short. In addition, different search engines perform our search string on different scales, although we have tried to limit the search string to the Title, Abstract, and Keywords fields. Thus, we define three pre-exclusion criteria (PEC) to eliminate such papers to relieve the subsequent workload. Due to the utilization of Google Sheets for recording papers, these straightforward exclusion criteria can be

automatically applied through functions. It is worth noting that for PEC3, we employed regex matching for the title, abstract, and keywords of each paper against the predefined search string. After systematic filtering, 1107 papers were left.

–PEC1 The length of the paper is less than six pages.

–PEC2 The paper is not written in English.

–PEC3 The combined content of the Title, Abstract, and Keywords in the paper does not match our search string.

3.2.3. Three-round selection

All the papers that pass the automated filtering are then manually evaluated for relevance. The overall selection criterion we defined is that the paper should propose an approach that primarily uses machine learning to assist RE-related tasks.

Inclusion and exclusion criteria. To better enforce the selection criterion, we have decomposed it into a series of inclusion criteria (IC) and exclusion criteria (EC) shown in Table 2. If the paper meets any exclusion criteria, we exclude it; and only if it meets all the inclusion criteria, we will include it.

Selection protocol. Two authors reviewed each paper regarding the inclusion and exclusion criteria. We first conducted a pilot study on 40 papers picked from the dataset randomly. If there were conflicting opinions, the two authors would explicitly express their concerns and resolve them. We used this pilot study to improve our criteria and enable the two authors to agree on the practical details. After reaching a consensus on the selection of the papers, the two authors selected the remaining papers. If there were any conflicts that the two authors could not resolve, we would bring in a third researcher for additional review.

Selection process. As shown in Fig. 1, we divide the manual inspection process into three rounds. In Round 1, reviewers read only the title and abstract of each paper to decide whether to exclude it. We move all the remaining papers to the next round. In Round 2, reviewers read both the introduction and conclusion of all the remaining papers. Same as before, we retain these papers until the next round, except for those excluded papers. In Round 3, reviewers need to read the full text and then make the final decisions. After the three-round manual selection, we selected 203 papers.

3.2.4. Domain knowledge supplementation

Machine learning, a widespread technique in recent years, has no shortage of applications in RE. Over the years, our research focus has centred on the ML4RE domain. Specifically, we have collected and

Table 2
Inclusion and exclusion criteria.

Aspects	Inclusion criteria		Exclusion criteria	
Type	IC0	Original proposal	EC0	Empirical study, evaluation
Goal	IC1	The purpose of the paper is to improve the certain stage or activity of <i>Requirements Engineering</i>	EC1	The paper targets the general software engineering or the whole life cycle of software development rather than the <i>Requirements Engineering</i> specifically
Method	IC2	The main methods employed by the paper are <i>Machine Learning</i> techniques	EC2	The paper is about developing/designing <i>Machine Learning</i> applications/systems
Reviewed method	IC3	Peer-reviewed journal papers and conference & workshop papers	EC3	Non-peer-reviewed papers such as those published in magazines, notes, etc.

summarized relevant papers from typical RE venues, including RE, REJ, REFSQ, and AIRE, every year since 2010. We selected the papers not among the 1845 papers and conducted a three-round selection for these papers. After this step, we obtained a total of 217 papers, with 14 new papers.

3.2.5. Snowball search

To make our study more comprehensive, we conducted a snowball search by scanning the references of the papers selected via the previous procedure.

References extraction. We extracted about 9000 references from all 217 papers in the first round of our snowball search.

References filter. We first filtered the 9000 references automatically and systematically. Specifically, we took the references that matched the search string in Table 1. After that, we obtained 244 references. Before we extracted the paper titles from these references manually, we removed all the papers before 2010 and less than six pages. We then removed the duplicate references, leaving us 10 unique items as candidates for the next procedure.

Three-round selection. This three-round selection procedure is the same as in Section 3.2.3. After three-round selection, one single paper was selected, and repeated snowballing on this paper did not yield any new papers. In other words, we had a total of 218 papers after the snowball search.

3.2.6. Extended work identification and filter

We found that 11 pairs had extended relationships in the 218 papers. We considered the extended articles as the outcome of these research works. Therefore, we excluded the pre-expansion papers. Eventually, in the white part, our final dataset contains 207 papers.¹ We identify these papers by their paper numbers in the following analysis.

3.3. Grey part selection process

Fig. 2 presents the procedure of the selection process in the grey part and the number of questions derived at each step is also in the figure. The following three subsections present the details of each step.

3.3.1. Search source selection

According to Garousi et al.'s classification [39], Stack Overflow is considered a 2nd tier grey literature source with moderate credibility, allowing for quantitative analysis. We selected Stack Overflow as a data source in the grey part due to its accessible and analyzable nature.

¹ The list of the 207 papers in the white part: <https://doi.org/10.5281/zenodo.8219116>.

Table 3
The tags used in the grey part.

Category	Keywords
RE tags	requirements, user stories, system requirements, requirements management, tweets, minimum requirements, user feedback, backlog, tfs, jira, ibm doors, ms word, cucumber, enterprise architect, ibm rational, fitness, mks integrity, rally, sysml, azure devops, trello, doxygen, confluence
ML tags	machine learning, supervised learning, nlp, unsupervised learning, neural network, decision tree, deep learning, svm, con vneural network, recurrent neural network, lstm, reinforcement learning, regression, word2vec, bert language model, tensorflow, pytorch, bayesian networks, scikit learn, transformer model, encoder decoder, named entity recognition, roberta language model, huggingface transformers, huggingface trainer, shap, lda, large language model, embedding, topic modeling, langchain, spacy3, spacy, vector search, seq2seq, distilbert, sentence transformers, artificial intelligence, gnn, openai api, mobilenet, pytorch lightning, tensorflow2.0, tensorflow lite, text classification, knn, kaggle, multiclass classification, streamlit, keras, tf.keras, text mining

3.3.2. Tag-based filtering

We found a total of 23,758,292 questions on the Stack Overflow. We utilized appropriate tags as categorization tools to identify and filter questions related to requirements engineering. Specifically, we selected six tags, they are: *requirements*, *user-stories*, *system-requirements*, *requirements-management*, *minimum-requirements*, and *REtags & MLtags*, where the contents of REtags and MLtags are detailed in Table 3. We then utilized Stack Overflow's API to extract all questions with any of these tags from 2010 to 2022. This endeavour resulted in respective question counts of 530, 183, 62, 31, 22 and 76.

Duplication Removal. In Stack Overflow, questions can be associated with multiple tags, potentially encompassing more than one of our selected tags. After the duplication, there are 861 questions left.

3.3.3. Manual selection

Our focus is on practical challenges faced by RE and deriving practical requirements from them. Therefore, all 861 questions that successfully passed tag-based filtering are then manually evaluated for their relevance to RE. Two authors evaluated each question in terms of the selection criterion. An initial pilot study involving 50 randomly chosen questions was conducted to address conflicting opinions, allowing the authors to reach a consensus. Subsequently, the two authors evaluated the remaining questions. Eventually, 375 questions were selected for the grey part.²

3.4. Data extraction and synthesis

This section describes the data extraction and synthesis procedures for the white, grey, and joint parts. We identified a set of tags outlining

² The list of the 375 questions in the grey part: <https://doi.org/10.5281/zenodo.10977763>.

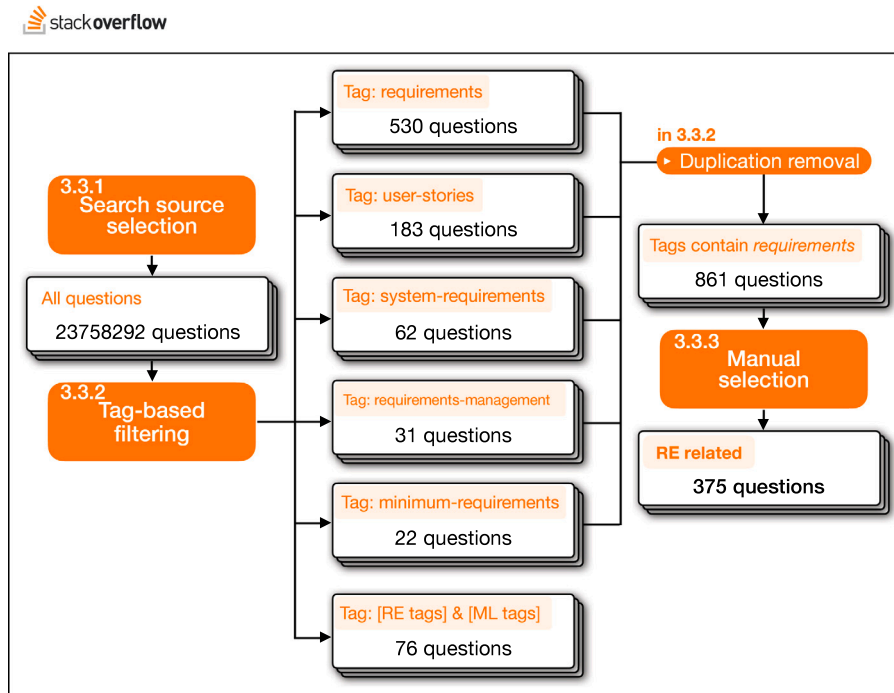


Fig. 2. The selection process of the grey part questions.

the information to be extracted from the search results. These tags function as metrics, aiming to answer the RQs based on the acquired information.

3.4.1. White data extraction and synthesis

To address RQ1, we intend to extract four aspects of information from the papers, each encompassing a set of tags as illustrated in Fig. 3. The descriptions of these four aspects and their corresponding tags are outlined below.

Basic information. This aspect is designed to understand the basic situation of the papers, including four tags: *Publication years*, *Publication venues*, *Publications types*, and *Citations*.

Problem domain. This aspect aims to understand the academic community's focus on requirements engineering, including two tags: *RE activities* and *RE tasks*. For *RE activities*, according to the software engineering body of knowledge (SWEBOOK), we classify each paper into one of five activities: *requirements elicitation*, *requirements analysis*, *requirements specification*, *requirements validation* and *requirements management*. For *RE tasks*, we extract the specific RE task from each paper according to its description.

Solution domain. This aspect is to investigate the usage of machine learning methods in ML4RE, including three tags: *ML tasks*, *ML techniques*, and *ML tools*. For *ML tasks*, we extract ML task types from each paper, e.g., classification, and clustering. Then, we extract particular ML techniques and ML tools used in each paper. For *Reproducibility*, we record the reproduction material types provided in each paper: *data*, *model codes*, and *experiment artifacts*.

Data. This aspect is to study the data analysed in ML4RE, including two tags: *Data types* and *Data sources*. For *Data types*, we initially categorize the data into one of four general types and then record the specific data based on the content of the paper. The four general types of data are: *requirement artifacts*, *user voices*, *domain data*, and *artifacts in SE*. For *Data sources*, we classify into one of five categories: *public*, *crawled*, *private*, *original*, and *no mention*. In addition, we extract the specific data sources.

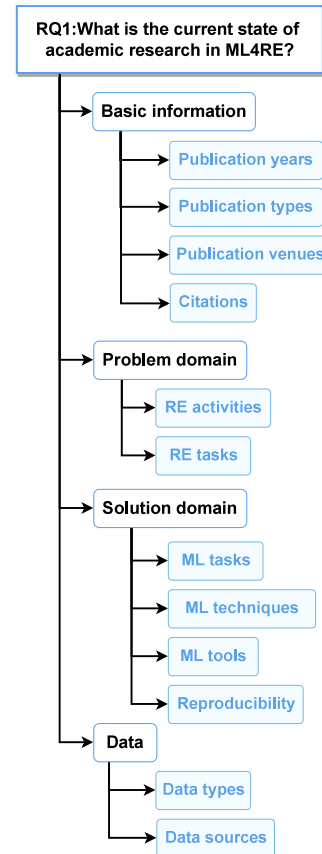


Fig. 3. Tags in the white part.

3.4.2. Grey data extraction and synthesis

To address RQ2, we designed to extract four aspects of information from questions and answers, each encompassing a set of tags as

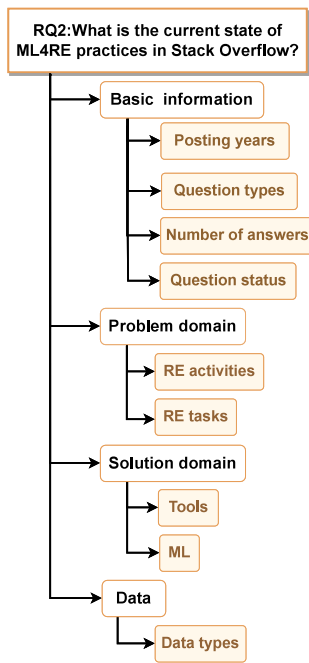


Fig. 4. Tags in the grey part.

illustrated in Fig. 4. The descriptions of these four aspects and their corresponding tags are outlined below.

Basic information. This aspect is designed to understand the basic situation of the questions, including four tags: *Posting years*, *Question types*, *Question status*, and *Number of answers*. For *Question types*, based on the content, we categorize questions into five types: *conceptual definitions*, *methodologies in practice*, *specific problems in practicing RE*, *problems during using tool*, and *need for solutions*. For *Question status*, we record whether the question asker accepted the answer for each question.

Problem domain. This aspect aims to study the practical challenges of requirements engineering faced in the Stack Overflow community, including two tags: *RE activities* and *RE tasks*. Similar to the white part, we classify each question into one of the five RE activities according to SWEBOK. Furthermore, we prioritize the tasks that were identified in the white part while assigning RE tasks to the questions.

Solution domain. This aspect is to investigate the usage of tools in the Stack Overflow community, including two tags: *Tools* and *ML*. We record the RE tools mentioned in the questions and answers respectively. For *ML*, we extract ML-related information following the process in the white part.

Data. This aspect is to study the data analysed in the Stack Overflow community, including only one tag: *Data types*. First, we record the specific types of data processed in each question. Then, we group all specific data types into more general data types that are the same as those in the white part.

3.4.3. Joint analysis data synthesis

To address RQ3, we intend to compare four aspects of information from the white part and grey part, each encompassing a set of tags as illustrated in Fig. 5. The descriptions of these four aspects and their corresponding tags are outlined below.

Basic information. This aspect is designed to compare the basic situation between academic research and practical applications on Stack Overflow, including one tag: *Trend*. Specifically, *Trend* is to discern the focus of both academia and practices in Stack Overflow by comparing *Publication years* from the white part and *Posting years* from the grey part.

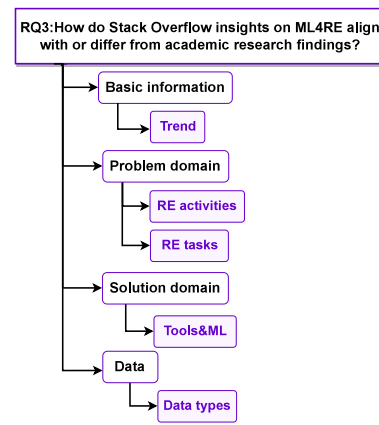


Fig. 5. Tags in the joint analysis part.

Problem domain. This aspect aims to compare the emphasis on requirements engineering between two parts, including two tags: *RE activities* and *RE tasks*. We conduct a comparison based on *RE activities* and *RE tasks* extracted earlier in the white part and grey part.

Solution domain. This aspect is to compare the methods employed by academia and practices in Stack Overflow in addressing RE problems, including one tag: *Tools&ML*. We conduct this comparison based on *ML techniques* extracted in the white part versus *Tools* and *ML* in the grey part.

Data. This aspect is designed to compare the data examined in academic research with the data types discussed and analysed within the Stack Overflow community, including one tag: *Data types*. The comparison is based on the *Data types* extracted earlier in the white part and grey part.

Similar to the previous manual selection, we assigned two researchers to conduct data processing for each paper to ensure reliability.

4. White part results (RQ1)

Following the design in 3.4.1, we present the information extracted from papers for answering RQ1 with some carefully designed visualizations, as [37] recommended.

4.1. Basic information

In this subsection, we provide some basic information to understand the situation in the academic community.

4.1.1. Publication years

In the white part, we have selected 207 papers. Fig. 6 shows the number of papers on ML4RE per year from 2010 to 2022. As we can see from the figure, research on this topic has grown as a quadratic function over the last decade, and the number of papers has increased significantly from 2016. To facilitate the later discussion, we tentatively classify the last 13 years into two phases: the initial phase (2010–2015) and the developing phase (2016–2022).

4.1.2. Publication types

There are 115 conference papers, 66 journal papers, and 26 workshop papers in the papers we selected. The number of each type of paper per year is in Fig. 7. Specifically, the conference papers number rose rapidly in the first four years (2016–2019) after entering the developing phase. As research work accumulates, the number of journal papers has started to surge in 2020. During the five years before 2020, there were only an average of 4.6 journal papers per year. However, in the 2020 year alone, 13 journal papers were published in this field. In 2022, they published 17 journal papers, reaching new heights.

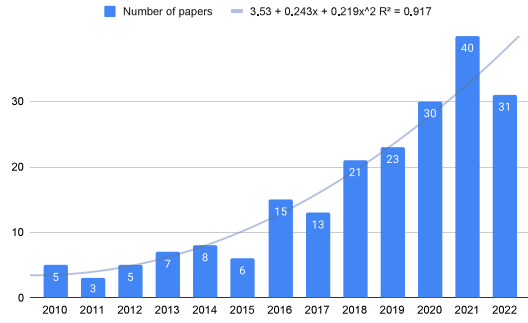


Fig. 6. The number of papers on ML4RE per year from 2010 to 2022.

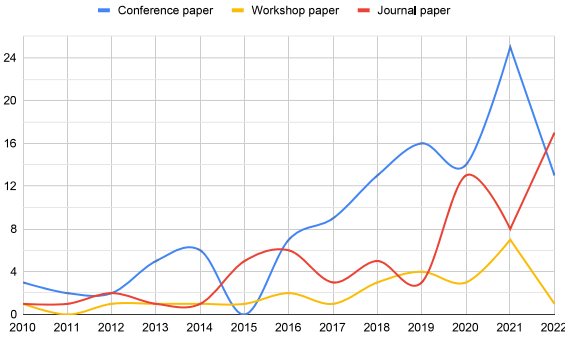


Fig. 7. Number of papers per year in each types.

4.1.3. Publication venues

Researchers published 207 papers we selected in 129 venues. Table 4 shows the overview of publication venues, with venues that appear only once categorized as *others*. As Table 4 shows, the 24 venues with multiple publications have nearly half of the papers (102/207, 49.3%) published. They are almost all classic venues in the RE community. The remaining 105 papers appeared in a wide range of venues (105/207, 50.7%), which helps share ML4RE beyond the RE community.

In our survey, we found that 16 papers mention extended relationships. Table 5 gives information on these 16 pairs of papers with extended relationships. The vast majority of them are cases in one conference paper extended to one journal paper. For example, there were five journal papers extended from conference papers in 2020. This result further indicates that journal articles stem from the accumulation of previous research work in the ML4RE field.

4.1.4. Citations

We have documented the citation relationships among selected papers, recording the frequency of each paper being referenced by other chosen works. Ultimately, we have identified papers with the highest citation counts: P053, P002, P022, P014, P018, P040, P058, P023, and P051.

The significant citations these papers amassed resonate with their representative role in the developmental stages of requirements engineering. Specifically, P002 (2010) employed naive Bayesian techniques to autonomously detect and classify non-functional requirements. Subsequent studies, namely P014, P018, P022, and P023 (2013–2014), delved deeper into the application of natural language processing techniques, such as topic modeling. Later, P040, P053, P058, and P051 (2016–2017) shifted their focus towards leveraging deep learning and text mining for automated analysis of vast datasets. In essence, the substantial citations of these papers may stem from their foundational role in laying the groundwork for subsequent research at each stage.

Table 4

Publication venues.

Paper type	Venue	Count
Journal papers	Appl. Sci.	4
	ESE	3
	IEEE Access	3
	IET Software	2
	IJACSA	2
	IST	4
	JSS	5
	REJ	7
	SQJ	2
	SoSyM	3
	TSE	3
	Others	28
Conference papers	APSEC	2
	ASE	3
	COMPSAC	2
	EASE	3
	ICIT	2
	ICSE	6
	MODELS	3
	RE	26
	SERA	2
	WorldCIST	2
	Others	64
Workshop papers	AIRE, co-located with RE	9
	RAISE, co-located with ICSE	2
	REW, co-located with RE	2
	Others	13

Table 5

Extended papers information.

No.	Before extension			After extension		
	Year	T	Venue	Year	T	Venue
1	2010	C	RE	2011	J	REJ
2	2015	C	RE	2015	J	REJ
3	2015	C	RE	2016	J	REJ
4	2017	C	RE	2017	J	REJ
5	2018	C	WI	2019	C	ICTAI
6	2018	C	APSEC	2020	J	JSS
7	2019	C	ICSR	2020	J	JSS
8	2019	C	REFSQ	2020	J	EMSE
9	2019	C	RE	2020	J	EMSE
10	2019	C	RE	2020	J	J SOFTW-EVOL PROC
11	2020	C	RE	2020	C	MODELS
12	2020	C	PoEM	2021	J	SoSyM
13	2018	C	RE	2021	J	ESE
14	2020	C	RE	2021	J	TSE
15	2021	C	RE	2022	J	PR
16	2020	C	RE	2022	J	RE

T = Type, C = Conference paper, J = Journal paper.

4.2. Problem domain

In this subsection, we present the academic community's focus on RE.

4.2.1. RE activities

We counted the number of RE activities assisted by ML techniques. Requirements analysis is the most supported activity by ML techniques, with nearly half of the papers (90/207, 43.48%). For the rest, the number of requirements elicitation and validation is similar, 59 for the former one and 36 for the latter one. They accounted for over 40% of all 207 selected papers together. At last, 22 studies (10.63%) used ML techniques in the requirements management phase. No study used ML techniques for the requirements specification in our selected papers.

4.2.2. RE tasks

We summarize the RE activities and corresponding RE tasks in Fig. 8. The 207 papers focus on 42 different tasks. Requirements

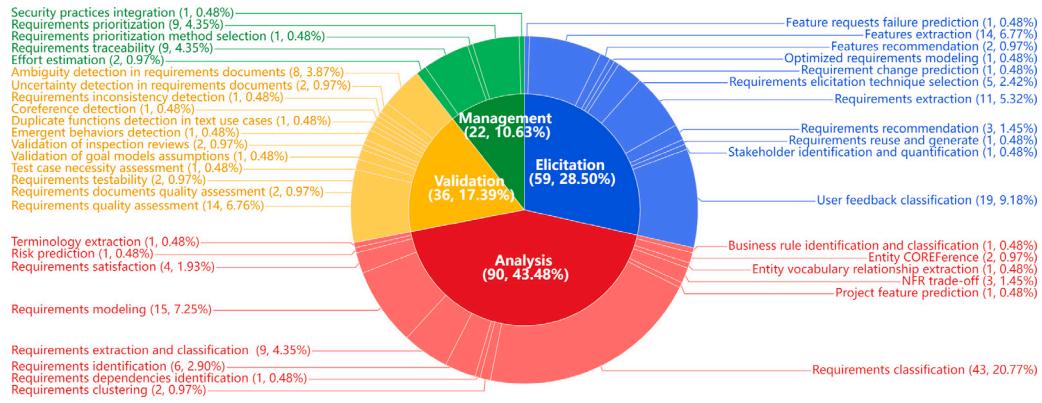


Fig. 8. Requirements activities and corresponding tasks.

classification (43/207, 20.77%) is the most explored task during the requirements analysis and even in all the RE activities. Requirements modeling (15, 7.25%), requirements extraction and classification (9, 4.35%), and requirements identification (6, 2.90%) are also popular tasks in the requirements analysis process. As for the elicitation phase, user feedback classification (19, 9.18%), features extraction (14, 6.77%) and requirements extraction (11, 5.32%) are the tasks that received the most attention. Requirements quality assessment (14, 6.76%) and ambiguity detection in requirements documents (8, 3.87%) are the most explored tasks during the requirements validation process. During the requirements management, requirements prioritization (9, 4.35%) and requirements traceability (9, 4.35%) are the focus of the researchers' attention.

The eleven tasks mentioned above are the ones that have received the most attention from researchers in all phases of RE. 157 papers corresponded to these eleven tasks, representing 75.85% of the total (157/207). It is worth noting that the remaining 50 papers dealt with 31 research tasks scattered across the various stages of requirements engineering. This data indicates that a large number (31/42, 73.81%) of RE tasks in the ML4RE domain still require further investigation.

4.3. Solution domain

In this subsection, we present the situation of ML techniques employed in addressing RE problems in the academic.

4.3.1. ML tasks

We counted the types of tasks that ML techniques processed each year, as shown in Fig. 9. It is evident that classification constitutes a significant portion, while other types of tasks have a relatively small share.

4.3.2. ML techniques

In our selection of 207 papers, researchers used 72 different ML techniques 308 times. Fig. 10 shows the rank of these techniques by the number of uses. We classify all 47 ML techniques that are used only once as *Others*. The most frequently used ML technique is SVM (used 37 times), while the next most used techniques are CNN (20 times), Decision Tree (17 times), BERT (17 times), and Random Forest (15 times). These data reveal a significant trend in the use of ML techniques in ML4RE papers. High-frequency utilization of advanced methods such as BERT indicates a sustained exploration of cutting-edge technologies by researchers. Additionally, the continued effectiveness of traditional machine learning approaches like Naive Bayes is evident from their consistent usage.

4.3.3. ML tools

More than half (123/207) of the researchers mentioned ML tools they used in their papers, they used 31 ML tools 120 times. The majority of ML tools are scikit-learn, followed by Weka, TensorFlow, Keras, Pytorch, and then Matlab.

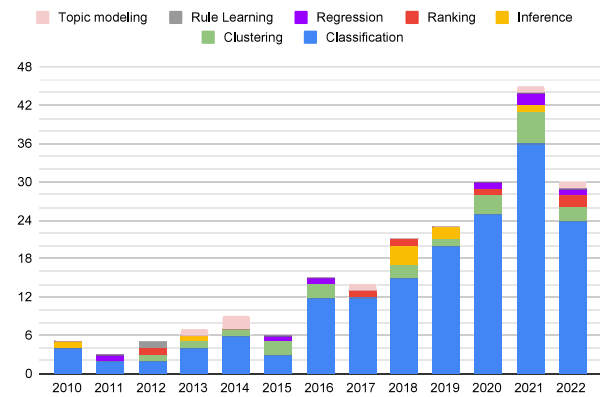


Fig. 9. Types of tasks that ML techniques processed per year.

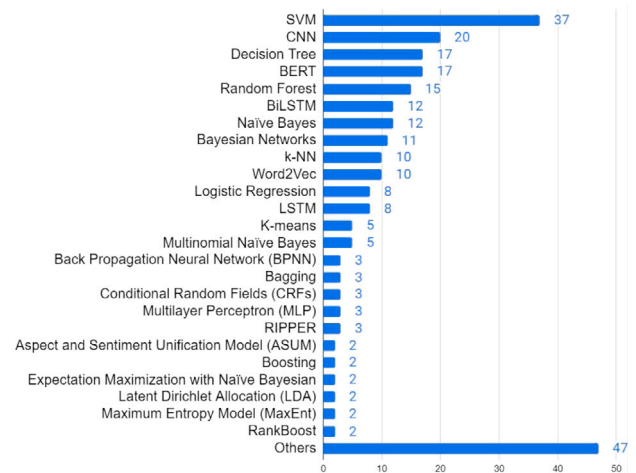


Fig. 10. Ranking of frequency of use of ML techniques.

4.3.4. Reproducibility

As for reproducing research works in the ML4RE field, researchers provide materials in three aspects: data, model codes, and empirical experiment artifacts. Close to forty percent (77/207, 37.2%) of the papers provided at least one material for reproduction. The majority of reproduction materials are found in the data, followed by model codes, with experiment artifacts being the least. We also documented

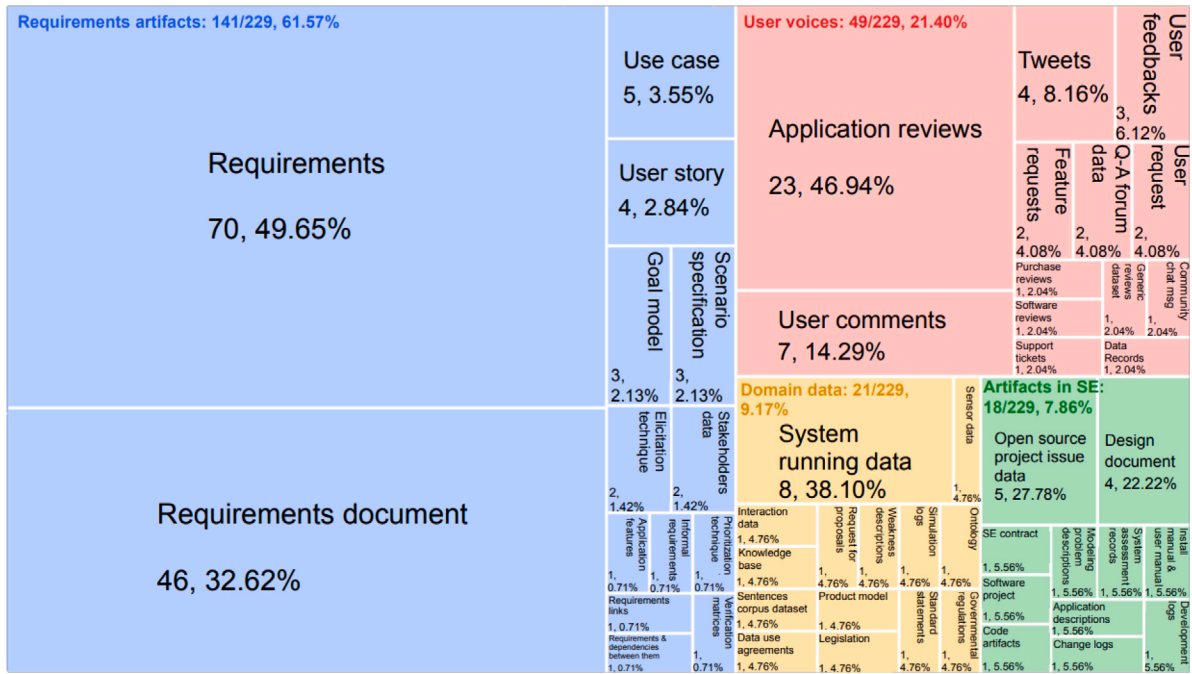


Fig. 11. All data types.

the materials provided in the papers with the aim of aiding researchers in accessing additional resources.³

4.4. Data

In this subsection, we introduce the situation of RE domain data processed by academia.

4.4.1. Data types

By scrutinizing the types of data used by ML techniques, we identified 53 types and grouped them into four general types, as shown in Fig. 11. We see that over 60% of the data falls into the requirements artifacts, which is the most intuitive data that needs to be processed to support RE activities and tasks. Of the remaining data, more than half are user voices, accounting for 21.40% of the total data. This data includes user feedback, requests, and user reviews of software and applications, together with messages from online forums and Twitter. User reviews make up the majority of the user voices. Other remaining data include artifacts in SE (7.86%) and domain data (9.17%). Artifacts in SE include the data from the software development life cycle. It is worth noting that here the artifacts in SE do not include requirements artifacts as they belong to the separate requirements artifacts category. Domain data includes a wealth of domain knowledge and is varied.

4.4.2. Data sources

We recorded the sources of data used, as shown in Fig. 12. Sixteen papers were vague about the source of all or some of the data they used, without explicitly mentioning the specific source of data they used. In the initial phase, papers using public data are more than papers using data from other sources. This predominance of public data is still evident in the developing phase. Furthermore, data from crawled and private have shown a clear upward trend from 2018.

Next, we describe the specific data source in detail in order of public, crawled, private, and original.

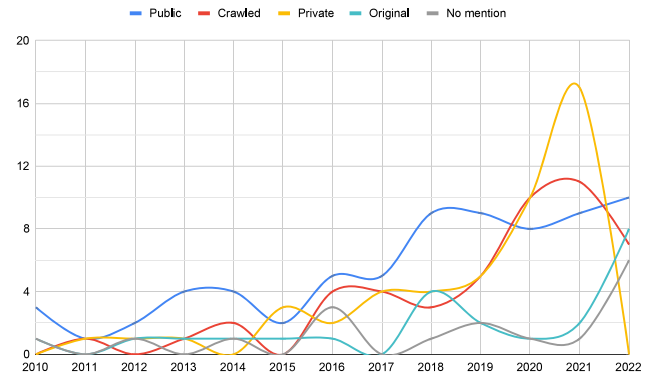


Fig. 12. All data sources by year.

Public data. We define public data as accessible datasets that are publicly available. We found that 76 papers used 49 public data sources 103 times. Several classic public datasets have been extensively used in the ML4RE domain. For instance, the PROMISE dataset was employed in 25 papers, making it the most frequently used dataset among our selected papers. Additionally, the RE17 data challenge was utilized in 6 papers. Besides these, other notable public datasets include PURE, iTrust, RE@UTS, and tera-promise, among others.

Crawled data. We specify crawled data as data acquired through web crawling. The most common place to crawl data is the Application store (19/58, 32.8%), with the Google Play Store and Apple App Store being the most common data sources. Due to the rapid development of mobile application stores/markets, many application reviews and descriptions have been available, which is one of the reasons for the increase of papers using crawled data in the developing phase.

Private data. If one paper explicitly mentioned that their data came from real industrial projects or other practical scenarios, we record the data source as practical. If researchers say that their data came from projects, we record such data sources as the project type. Based on our analysis, 37 papers utilized practical data, 12 papers used project data, and 4 papers did not specify their data sources.

³ The reproduction materials provided in the white part: <https://doi.org/10.5281/zenodo.10130209>.

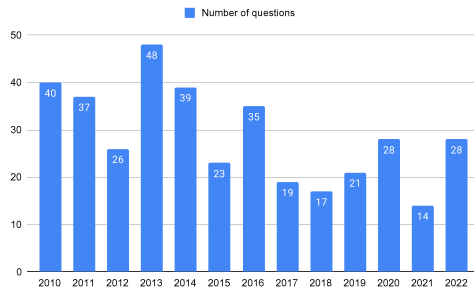


Fig. 13. The number of questions on Stack Overflow per year from 2010 to 2022.

Original data. We define data belonging to researchers themselves as original data. Specifically, original data can come from system running processes or can be collected from experiments.

4.5. The state of the ML4RE in academia

In summary, the current landscape of ML4RE research reveals a consistent growth in paper quantity and a shift toward higher-quality journal publications. In the realm of RE activities, there is a predominant focus on RE analysis, with a diverse range of specific tasks explored. Moreover, ML techniques such as SVM, CNN, decision trees, and BERT are widely used, and some work also provides materials to support open source. The data types addressed are diverse, with an emphasis on requirements artifacts, and the sources are mainly public data.

5. Grey part results (RQ2)

Following the design in 3.4.2, we present the information extracted from selected questions for addressing RQ2.

5.1. Basic information

In this subsection, we provide some basic information to understand the situation in practices in Stack Overflow.

5.1.1. Posting years

After the search and filtering process in Section 3.3, we selected 375 questions in the grey part. Fig. 13 shows the number of questions related to RE on Stack Overflow between 2010 and 2022. The number of RE-related questions on Stack Overflow has been declining overall. In 2010, 2011, 2013, 2014, and 2016, there were more than 30 questions each year. From 2015 to 2022, there were around 20 questions each year.

5.1.2. Question types

Depending on the content of the questions, we define the question types here as *conceptual definitions*, *methodologies in practice*, *specific problems in practicing RE*, *problems during using tools*, and *need for solutions*. Example questions of these types are shown in Table 6. Questions about specific problems in practicing RE were the most frequent (36.0%), followed by problems in using tools (24.4%). Methodologies in practice came in third (21.8%), while need for solutions had 10.5% of the questions. Conceptual definition had only 7.3% of the questions.

Based on the content of the questions practitioners asked on Stack Overflow about RE, the three types of questions, conceptual definitions, methodologies in practice, and specific problems in practicing RE, are all related to domain knowledge of RE. It suggests that there are difficulties in applying theories in RE to practice. From the remaining two types of questions, we can find out those RE activities, tasks and requirements where practitioners are currently using tools.

Table 6

Question types and examples.

Question type	Example
Conceptual definitions	Do any software developers know what the General Systems Model or an Organizational Model is?
Methodologies in practice	What were the requirement elicitation techniques before use cases?
Specific problems in practicing RE	How to organize information about program solution?
Problems during using tools	What is the best way to link formal specs to JIRA enhancement requests?
Need for solutions	Which tool to use to receive customer requirements?

5.1.3. Number of answers

We systematically statistic the answers to the 375 questions from Stack Overflow. Only 21 of the 375 questions were left unanswered. All these questions received a total of 665 answers, with an average of 1.8 answers per question.

5.1.4. Question status

Among the 375 questions, most of the questions (354) received answers, with 196 questions having accepted the answers. The remaining 179 questions, not accepted, may suggest that the askers are either dissatisfied with the current responses or are awaiting more satisfactory solutions.

5.2. Problem domain

In this section, we introduce the RE activities and tasks mentioned in each question type to demonstrate the Stack Overflow community's attention to RE.

Concepts definitions. Questions concerning conceptual definitions address the definition of one or more specific RE concepts. Specifically, 27 questions mentioned a total of 16 RE concepts 41 times. Eleven of these concepts were mentioned only once, accounting for 26.83% of the total mentions(11/41). The most frequently mentioned concepts are functional requirements (FR), use cases, and requirements. FR appeared in eleven questions, use cases, and requirements in six separately. It is worth noting that six questions directly asked about the definition of non-functional requirements.

According to our findings, we subdivide conceptual definition questions into two categories: questions that address the definition of a particular concept, and ones that distinguish between two or more concepts. Of these, questions that distinguish between concepts are more frequent, containing 22 questions, accounting for 81.5% (22/27) of the total number of conceptual definition questions. The most frequently differentiated concepts are FR and NFR, with five questions distinguishing these two concepts. In addition, three questions distinguish between FR and Use case. There are five questions that address the definition of four different concepts: logical requirements, requirements types, non-functional use cases, and non-interactive functional requirements.

Methodologies in practice. There are 83 questions asked about RE methodologies involving 24 RE activities. Of these, nearly one third (30/83, 36.1%) involved analysis, followed by specification (24/83, 28.9%) and management (22/83, 26.5%). The number of questions involving elicitation and validation activities was relatively close, with 5 and 2 questions, respectively.

For RE tasks, we introduce the first few. The most common task is writing requirements in the specification activity, with a total of 14 questions. Next is specification creation and behaviour-driven requirements analysis, both with 8 questions. Then, it is followed by the requirements modeling, which has 7 questions.

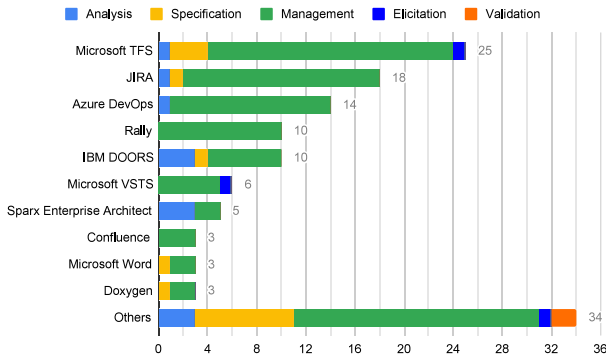


Fig. 14. Tools in questions.

Specific problems in practicing RE. There are 137 RE activities in the questions about specific problems in practicing RE. The RE activity with the most questions is the analysis activity, with 84 questions (72/124, 61.3%). The second most frequent is the specification activity with 33 questions.

For RE tasks, requirements modeling has the most questions, with 32 in total. Next, requirements classification has 14 questions, and writing requirements have 12 questions, while the other tasks in analysis and other RE activities have less than 10 questions each.

Problems during using tools. The most common RE activity is management, mentioned in 78 questions. Following that, specification, analysis, and validation are mentioned in 6, 6, and 2 question(s) respectively.

Because of the disparity between the number of questions in the management and the number of questions in the remaining four activities, we only count the specific tasks involved in the management activity. The most frequently addressed management task is requirements CRUD, with 21 questions related to it, accounting for nearly one third of all questions related to management activities. The second is requirements traceability, with 17 questions associated with it.

Need for solutions. The methods practitioners seek can be categorized into RE tools and ML techniques, collectively termed the solutions here. For RE activities, the most frequently mentioned is analysis, which is referred to in 16 questions. Next is the management, mentioned in 12 questions.

Then, we introduce the top four RE tasks. The task mentioned most by questions about tool requirements is requirements traceability, with 6 questions. Following that is specification creation, with 3 questions. The last two are requirements traceability and requirements document management, both mentioned in 2 questions.

5.3. Solution domain

In this subsection, we present the tools and ML employed to address RE problems in Stack Overflow community.

5.3.1. Tools

Tools in questions. We count all the RE tools mentioned in all 375 questions. A total of 80 questions (85/375, 22.7%) mentioned 40 tools 130 times. Fig. 14 shows the statistics, and tools that appear only once are marked as *Others*. The most frequently mentioned tool in the questions is Microsoft TFS, with 25 questions mentioned. Microsoft TFS is the tool used in RE Management activities. Following are Atlassian Jira and Azure DevOps, which are mentioned in 18 questions and 14 questions, respectively. They are both used for RE management activity. From the figure, we can find that most tools mentioned in the questions are used for RE management activities.

Tools in answers. We count all tools mentioned in all 665 answers. A total of 148 answers (148/665, 22.4%) mentioned 98 tools 206

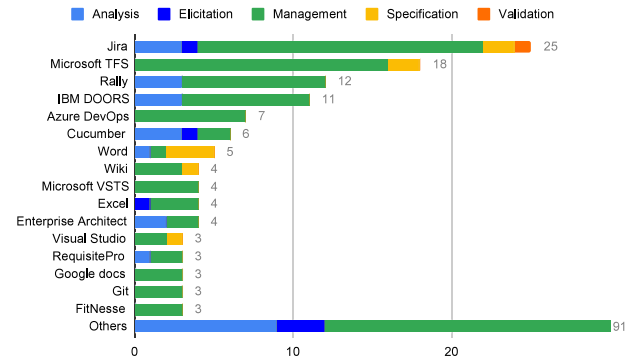


Fig. 15. Tools in answers.

times, an average of 1.4 tools per response (206/148). For each tool, we count the number of answers that mentioned it. Fig. 15 shows the results, and tools that appeared only once are marked as *Others*. The most mentioned tool in answers is Atlassian Jira, with 25 answers mentioned, and the following are Microsoft TFS and Rally, with 18 and 12 answers respectively. The majority of these top tools mentioned in the answers are utilized in RE management activity.

5.3.2. ML

ML tasks. We recorded the types of tasks addressed by the ML mentioned in the selected questions and corresponding answers, which consists of 35 questions and 28 answers. Among all the ML tasks mentioned, the most focus on classification, followed by clustering and topic modeling.

ML techniques. Among the 35 questions and corresponding answers we selected, 14 different ML techniques were mentioned 36 times. The most frequently mentioned ML technique are LDA, POS tagging and TF-IDF (5 times each), followed by neural networks and GSDMM (4 times each), and then k-NN (3 times). It can be observed that practitioners often utilize ML techniques related to text processing.

ML tools. Among the 35 questions and corresponding answers we selected, 13 different ML tools were mentioned 38 times. The most frequently mentioned ML tools are scikit-learn (8 times), followed by gensim (7 times), and then textblob and tm (4 times each).

5.4. Data

In this subsection, we introduce the situation of RE domain data processed by the Stack Overflow community. Additionally, it is important to note that, due to the less detailed mention of data in the questions compared to the white part, we only extracted data types here. From our analysis, all of the data involved in these questions are requirements artifacts and user voices. Specifically, the most frequently mentioned data is user story, accounting for 37.6% (141/375) of all. The remaining data types are mainly requirements, tweets, use case, and requirements documents.

5.5. The state of the ML4RE practices in Stack Overflow

In summary, the overall number of questions related to RE has seen a decline. While a substantial number of questions received answers, only a limited portion was accepted. The predominant focus in these questions revolves around analysis and management activities, with specific tasks like requirements modeling, writing requirements, and requirements traceability standing out. Frequently mentioned tools in the answers and questions include Microsoft TFS, Jira, IBM Rational DOORS, Rally, and Azure DevOps, while ML techniques comprise LDA, POS, TF-IDF, and neural networks. The majority type of data mentioned in the questions is requirements artifacts, with user stories being the most prevalent.

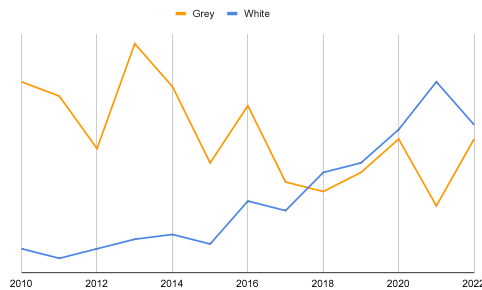


Fig. 16. Trends of publications in white and grey.

6. Joint part results (RQ3)

Following the design in 3.4.3, we give comparative results from the white and grey parts for answering RQ3.

6.1. Basic information

In this subsection, we compare the different trends of white part and grey part on ML4RE in Fig. 16.

Overall, there is an upward trend in the number of papers and a downward trend in the number of questions. Specifically, between 2010 and 2013, both the grey and white parts showed an upward trend. However, from 2014 to 2017, the grey part exhibited a declining trend while the white part showed an ascending trend. Subsequently, in the following years, the white part continued to gradually increase, whereas the grey part remained relatively stable.

6.2. Problem domain

In order to analyse the focus of academic research and real-world practices on the RE domain, we compare the RE activities and RE tasks in the white and grey parts.

6.2.1. RE activities

We compare the percentage of each RE activity involved in the white and grey parts. The RE analysis accounts for the largest proportion of both parts, about 40%. In contrast, the proportion of other RE activities shows a clear difference. Specifically, in the white part, none of the papers dealt with the RE specification. However, 19% of the questions in the grey part related to this activity. RE management is the second most common activity covered by the grey part, with 32% of questions. However, only 11% of the papers in the white part dealt with this. For RE elicitation and RE validation, the white part respectively account for around 26% and 17%. However, only about 5% and 3% of questions in the grey part relate to these two activities.

To specifically understand the focus on RE activities between the white and grey parts, we compare the trends of RE activities in these two parts, as shown in Fig. 17. Analysis stands out as the most emphasized activity in both. Elicitation and validation, while displaying an overall increasing trend in the white from 2010 to 2022, did not gain significant attention in the grey. On the other hand, management is noticeably prominent in the grey, exhibiting a stable trend over the years, whereas the white sector did not demonstrate substantial interest in this activity. Moreover, specification shows a declining trend in the grey, which did not get attention in the white part.

6.2.2. RE tasks

There are eight intersecting RE tasks between the white and grey parts: requirements modeling, requirements traceability, requirements

classification, requirements extraction, requirements quality assessment, requirements prioritization, requirements elicitation technique selection, and requirements reuse and generation. We rank these tasks based on their frequency of occurrence in the white and grey parts, allowing us to determine their relative attention in both areas. The tasks of mutual interest in both parts include requirements modeling and requirements classification. However, requirements quality assessment receives sole attention in the white part, while requirements traceability is the sole focus in the grey part.

6.3. Solution domain

We analyse the similarities and differences in methods taken by academic research and real-world practices in Stack Overflow in addressing issues in RE in this subsection.

In the white part, the most frequently used ML techniques are SVM, CNN, Decision Tree, Bert, Random Forest, and BiLSTM. Then, we compiled a comprehensive tally of tools that appeared in questions and answers, identifying the most frequently mentioned tools as Microsoft TFS, Jira, IBM Rational DOORS, Rally, and Azure DevOps. Additionally, some ML techniques were mentioned in parts of the questions and answers, primarily including LDA, POS, TF-IDF, and neural networks.

Therefore, it can be observed that the focus of the white part lies in traditional classification and regression methods, as well as deep learning models. On the other hand, the grey part's emphasis includes common software tools and some ML techniques relevant to text analysis and natural language processing.

6.4. Data

We compare the general data type and specific data type extracted from the white and grey parts to analyse their attention to the data.

The most commonly processed general data type in both the white and grey parts is requirement artifacts. We compare the usage of the top 6 specific data types within the requirement artifacts between the two parts, as shown in Fig. 18. In the white part, the most frequently used requirement artifacts are requirements and requirement documents, and these two are also commonly utilized in the grey part. Among the remaining data types in the grey part, the user story and use case received significant attention, especially the user story. However, in the white part, there is relatively less emphasis on these two types of data.

6.5. The similarities and differences between academic research and Stack Overflow practices

In summary, there is an increasing trend of academic papers while a decline in posted questions of the Stack Overflow community. Both parts prioritize RE analysis, requirements modeling, and requirements classification. However, academia emphasizes RE elicitation and validation, utilizing ML techniques like SVM, CNN, Decision Tree, and BERT. Conversely, the Stack Overflow community focuses more on RE management and specification, relying on tools such as Microsoft TFS, Jira, IBM Rational DOORS, Rally, and Azure DevOps, as well as ML techniques like LDA, POS, TF-IDF, and neural networks. For data, while both pay attention to requirements and requirement documents, the Stack Overflow community specifically emphasizes user story and use case, which receive less focus in academic research.

7. Discussion

In this section, we first present and analyse several interesting phenomena based on the previous results obtained from our research. Subsequently, drawing from these analyses, we present a set of research suggestions.

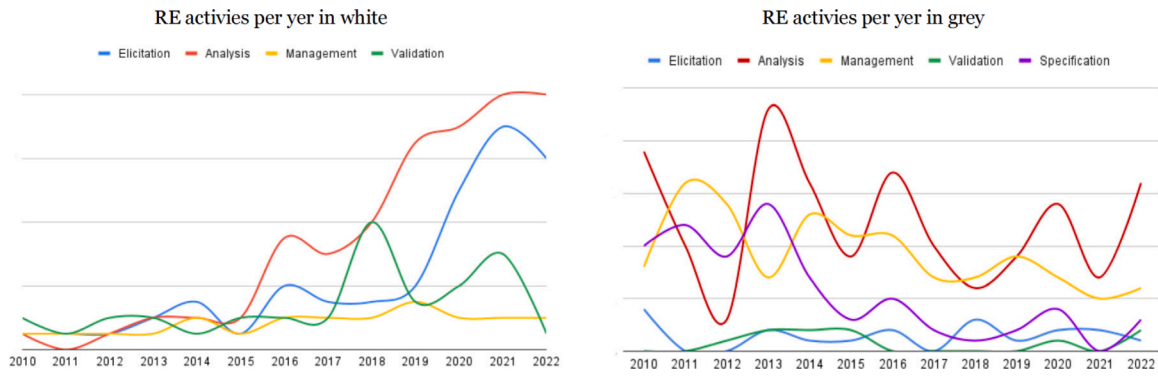


Fig. 17. The trends of RE activities per year in white and grey parts.

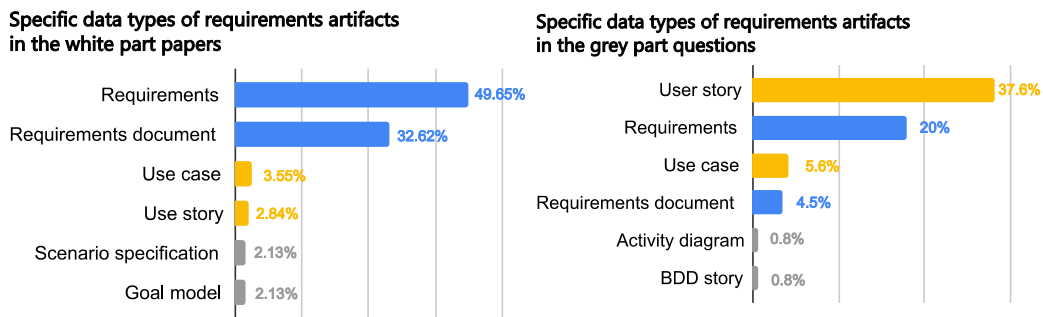


Fig. 18. The proportion of data type in white and grey parts.

7.1. Riding the wave of technological advancements

When analysing the selected papers, we notice significant trends and shifts closely linked to the advancement of technology. Firstly, we observed a significant increase in the number of ML4RE-related papers. This trend aligns with the rapid development of machine learning techniques and also correlates with the growth trend of the number of applications over the past 13 years [40,41]. As the number of applications increases, the demand for standing out in the competitive market becomes critical, making requirements even more crucial. Simultaneously, this surge has resulted in a large volume of analyzable data, with ML technology serving as an effective analytical method, driving the research in ML4RE.

The selected papers also follow a clear trajectory across different stages with the evolution of machine learning technology. We have summarized the roadmap as depicted in Fig. 19. In the early phase, spanning from 2010 to 2015, the focus primarily centred on traditional machine learning algorithms such as SVM and Bayesian methods. Subsequently, around 2016 to 2017, the research emphasis shifted towards foundational neural networks, like CNN, alongside word embedding techniques like word2vec. Then, from 2017 to 2019, there was a surge in the application of deep learning methods, notably the adoption of LSTM. Most recently, spanning from 2020 to 2022, attention gradually pivoted towards generative and pre-trained models such as BERT and GPT. It is anticipated that future research in ML4RE will continue to explore large language models extensively.

Moreover, although public data remains the largest source of data, the growth in research using such data has shown a slowdown in the past two years. Conversely, there has been a significant uptick in studies utilizing crawled data. One possible reason for this may be attributed to the era of big data and technological advancement. The omnipresence of digital technology in daily life generates copious amounts of data, facilitating researchers' study for more comprehensive and realistic data sources.

These observations in the ML4RE research response to technological advancements and provide valuable insights and directions for future studies.

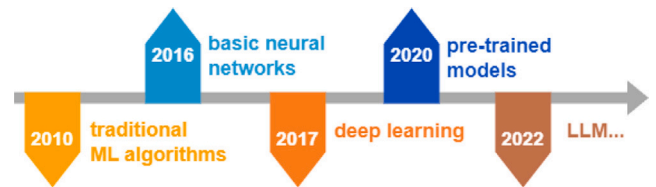


Fig. 19. The roadmap in ML4RE research.

7.2. Contrasting priorities between academic research and practices in Stack Overflow

In our comparative analysis of the academic research and practices in Stack Overflow in addressing issues in the RE domain, a distinct disparity emerges. Based on the findings, researchers tend to prefer advanced machine learning and deep learning (DL) techniques. Conversely, practitioners are observed to lean towards the utilization of existing software tools and natural language processing (NLP) techniques.

The underlying reason behind this disparity may lie in the academic emphasis on the innovation and complexity of technologies, whereas industry places greater importance on practicality and cost-effectiveness. ML and DL techniques utilized in the white part often entail high complexity and cost, requiring specialized knowledge and professionals for development and implementation. Additionally, these techniques typically demand substantial training data and computational resources. In contrast, the software tools and NLP techniques used in the grey part are generally more straightforward to implement and apply, as they have already been widely adopted in the industry and possess lower learning curves and costs.

Furthermore, only a portion of the questions and answers in the grey part mention ML techniques. This could be due to some questions revolving around defining the concepts, methods, and tools of RE. Therefore, focusing solely on RE domain knowledge is sufficient to

address these questions. Additionally, some questions may provide a specific scenario, with answers based on specific methods tailored to that scenario, without the need for ML techniques.

7.3. Disconnection between academic research and industrial practice

When comparing the focus on RE activities and data between the white and grey parts, we identified disparities that underscore a disconnection between academia and industry.

From Fig. 17, a disparity in attention within RE activities becomes evident. The grey part consistently raises some questions related to RE management, while the academic research in the white part does not show a marked increase in this domain. This difference might stem from inadequate communication between academia and industry, resulting in a lack of understanding within academia regarding the specific needs and directions in RE management.

Moreover, in Fig. 18, the grey part shows significant attention toward the user story and use case, aspects that receive relatively less research focus within academia. This lack of emphasis on these aspects in academic research also implies a potential disconnection between academia and industry. Furthermore, it might be associated with the sensitivity of these data regarding company privacy, making it challenging to acquire sufficient support and data for academic research.

Overall, while academia strives to explore novel methodologies, these approaches might not always immediately resonate with the practical problems and direct requirements of the industry. These disparities highlight the necessity for a closer alignment between research directions and real-world needs.

7.4. Suggestions

In this subsection, we plan to provide several suggestions based on our findings and discussions for researchers in the ML4RE field.

7.4.1. Constructing intelligent question-answering assistants

Following research on the grey part, it was found that questions about specific problems in practicing RE and conceptual definition account for 36.2% and 8.2%, respectively. This indicates that there are some questions that could be resolved by leveraging specialized knowledge in the field of RE. Given the current advancements in technology, developing intelligent question-answering assistants based on Large Language Models (LLM) is a promising research direction.

LLM represents a class of powerful NLP tools harnessing extensive pre-training and deep learning techniques. These models possess the capability to deeply understand, reason, and generate natural language text. This capability enables LLM to extract knowledge from vast textual data, understand context, and consequently engage in logical reasoning to generate accurate responses. Thus, constructing an intelligent question-answering assistant using LLM offers several distinct advantages: (1) a comprehensive repository of domain-specific knowledge stemming from LLM's extensive pre-training; (2) superior comprehension and reasoning abilities, allowing for precise resolution of complex RE problems; (3) real-time and detailed responsiveness in providing efficient and meticulous answers to questions.

Given these advantages, it is recommended to organize datasets from the RE domain and then fine-tune LLM toward developing an intelligent question-answering assistant. Such an assistant could effectively discern and resolve intricate concepts, offering tailored solutions for specific RE questions.

7.4.2. Bridging machine learning with industry tools

In light of our earlier discussions, it is apparent that the practitioners in the Stack Overflow community is inclined toward utilizing established tools such as Microsoft TFS, Jira, IBM Rational DOORS, Rally,

and Azure DevOps. However, as indicated by the statistical findings in Section 5.3.1, these tools are frequently referenced within the context of questions, presenting a 25% probability. This signifies the potential challenges and complexities practitioners face while using them. For instance, Question ID: 71746782, Title: "How to pass value of a field from Parent Work Item to Child in Azure DevOps?", illustrates the issues that may arise when utilizing these tools.

Hence, researchers are encouraged to develop intelligent adaptive tools that integrate machine learning methods. Leveraging the foundation of existing industry tools, these adaptive tools aim to address the challenges encountered in the industrial execution of requirements engineering tasks.

Furthermore, to advance the broader adoption of machine learning methods in industrial settings, researchers should actively support open-source initiatives. This involves contributing machine learning models, algorithms, and tools to a wider community. Our research uncovered that approximately 37 out of 207 selected papers provided model codes or experimental artifacts. Given the complexity of industrial practitioners reproducing code from academic papers, it is recommended that researchers translate innovative methods into useable tools, facilitating easier adoption within industrial settings.

7.4.3. Enhancing collaboration between academic research and industrial practice

Based on the discernible disparities in focus between academic research and the practices within Stack Overflow, we propose fortifying collaboration between these spheres to better align research endeavours with practical demands.

Our research reveals the heightened attention to RE management and user story by practitioners within Stack Overflow, areas that remain relatively unexplored within academic investigations. This discrepancy may stem from a disconnection between academia and industry, leading to a lack of understanding regarding the pressing needs of the industrial landscape. Thus, fostering close collaboration becomes pivotal. Conducting targeted research or surveys within industrial settings would enable researchers to delve deeper into genuine industry requisites, facilitating the identification of research directions. Simultaneously, collaborating with practitioners can furnish datasets from real-world scenarios, bridging the gaps in academic access to user story and ensuring research aligns with practical requirements.

Furthermore, the provision of more open datasets is crucial in the research community. It is noteworthy that while there are some papers that explicitly state the datasets they used, the accessibility of these datasets can become an issue over time. For example, the address of *tera-PROMISE*⁴ in the paper P096 we selected is currently inaccessible. A recommended practice is to share datasets on platforms like Zenodo and GitHub. These established platforms contribute to the long-term stability and accessibility of shared data.

8. Threats to validity

In this section, we analyse threats that might affect the validity of our work. Here we adopt a classification scheme from Runeson and Höst [42] that focuses on case study research in software engineering.

8.1. Construct validity

Construct validity reflects the extent to which the research protocol meets the research objectives and research questions. The representativeness of selected studies is the key threat to our study. For the grey literature review, we chose Stack Overflow as the information source. However, when searching using a combination of RE and ML keywords, the results were quite limited. We found that the questions retrieved

⁴ <https://terapromise.csc.ncsu.edu/#repo/view/head/requirements/nfr>.

using the RE keywords were mainly focused on specific scenarios, and thus, the corresponding answers were typically tailored to those specific scenarios. Although these answers seldom included content related to ML, we believe that these questions still help in analysing the challenges of real-world RE. Therefore, we decided to adopt two approaches for the grey part search: on the one hand, searching for questions directly related to RE, and on the other hand, combining the ML keywords with words that have some relevance to RE. We realize that this search method may not cover all the questions related to ML4RE, but we made our best effort to balance workload and content quality. We look forward to collecting more comprehensive grey resources in future research to further enrich our literature review.

8.2. Internal validity

Internal validity requires researchers to consider the factors that influence research as comprehensively as possible. If there is a factor that affects the quality of the study and we are not aware of that, there will be a threat to the internal validity. Our literature review design follows the guidelines of standard academic papers, which, to some extent, alleviates threats to internal validity. However, potential threats to internal validity still exist, particularly concerning the accuracy of manually labelled tags. During the process, a large amount of data needs to be counted and extracted. These tasks can be tedious and may result in inaccuracies or errors in recording data. To alleviate this issue, we created a shared spreadsheet, established labelling rules, and minimized the need for manual input, thus reducing the risk of errors by researchers. Additionally, we double-check the data to ensure its accuracy and utilize computational formulas whenever possible for statistical calculations. We believe this helps mitigate threats during the data collection process.

8.3. External validity

External validity reflects the ability to generalize the results of the study. One potential issue with external validity is the limited time frame of data. In our analysis, we utilized data from 2010 to 2022, including both white and grey literature, but did not consider studies published prior to 2010. Fortunately, the rapid advancement of machine learning since 2010 has helped reduce this concern. Another threat can be the limited scope of grey literature. Our primary contribution lies in providing an integrated perspective on grey literature. Stack Overflow is one of the most informative and suitable data sources, which best fits our needs in this paper. We made efforts to encompass a wide range of RE questions to mitigate the impact of Stack Overflow as the sole data source. However, it is essential to acknowledge that the generalizability of our findings may be influenced by the limitations of Stack Overflow. To address this, future work could gradually incorporate additional data sources to enhance the external validity of our study.

8.4. Conclusion validity

Conclusion validity concerns the reproducibility of the study. That means whether other researchers following the protocol of our paper and conducting the same study can obtain the same results. One threat could be related to bias in the data selection. To mitigate the threat, we have specified the inclusion and exclusion criteria for the data in both white and grey parts and have strictly enforced them. Two authors completed the data selection process and carried out a pilot study to align their views before starting. We introduced a third researcher to resolve the conflicts between the two. A similar threat exists in the subsequent data tagging process. Before we started processing the data, we clearly defined what data we needed to extract and how to extract it to mitigate this threat. As with the paper selection, two authors completed the process and followed the same conflicts resolution process

as in the paper selection. Besides, the researchers established a gold standard for labelling and used strict inclusion/exclusion criteria to unify differences among different researchers. In particular, the two researchers tagged papers simultaneously and periodically unified their tags, as the different terms extracted from different papers may refer to the same thing. In addition, the reliability of the mapping stage was essential to ensure that the interpretation of the data as objective and consistent with the original meaning behind the extracted data. We made several discussions and changes to the data presentation to ensure that the final charts reflected the original intent of the extracted data.

9. Conclusion and future work

In this paper, we have conducted a systematic literature review on ML4RE, integrating practitioners' insights from Stack Overflow. Toward academia, we systematically searched and selected academic publications from four academic databases (IEEE Xplore, ACM Digital Library, SpringerLink, and Scopus). We obtained 207 publications published in ML4RE between 2010 and 2022. For grey literature, we systematically searched Stack Overflow and obtained 375 questions asked by practitioners in RE practices from 2010 to 2022 with the answers to these questions. We analyse the literature's trends, RE activities, RE tasks, solutions, and data. By analysing and integrating the extracted information, we present the current status of ML4RE in academic research and practices in Stack Overflow. Lastly, we compare the results of these two parts to discern their similarities and differences. Based on our research results, we discuss some interesting phenomena worthy of attention in ML4RE and give some research suggestions.

Our future work could continue to investigate the latest research trends in this field. In addition, more grey literature could be involved to listen more fully to the voices of practitioners and the industry.

CRediT authorship contribution statement

Tong Li: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Xinran Zhang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Yunduo Wang:** Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Qixiang Zhou:** Writing – review & editing, Visualization, Investigation, Data curation. **Yiting Wang:** Writing – review & editing, Visualization, Data curation. **Fangqi Dong:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The link to the data is specified in the paper.

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